

ANIK SEN

+886 926729283 | +60 11 126 79199 | aniksen360@gmail.com | Tainan, Taiwan

 LinkedIn |  GitHub |  Google Scholar |  Website

ABOUT ME

I am working as a visiting research trainee at NCKU, Taiwan. I have completed my work completion defence of my **Master of Computing** (By Research) programme at **Multimedia University**, Melaka, Malaysia. I worked on an FRGS-funded research project as a **Graduate Research Assistant** in the past two years. I completed my Bachelor of Science in **Computer Science & Engineering**, specializing in Software Engineering, at **American International University-Bangladesh (AIUB)**. My research interest areas are **Information Security, Machine Learning, Federated Learning, and Cryptography**.

WORK EXPERIENCE

- **Visiting Research Trainee (NSTC, IIPP)** 06/2026 - Current
ISS-NET Lab, National Cheng Kung University (NCKU) Tainan, Taiwan
Responsibilities include:
 - Serving as the project lead and assisting undergraduate students.
 - Contributing to ongoing lab projects related to Machine Learning (ML) and Federated Learning (FL).
- **Graduate Research Assistant** 05/2024 - 05/2026
Multimedia University Melaka, Malaysia
Responsibilities include:
 - Conducting research on privacy-preserving machine learning to secure data sharing in Federated Learning settings.
 - Integrating cryptographic techniques, including Homomorphic Encryption, to enhance data privacy.
 - Exploring applications of Generative AI and Large Language Models (LLMs).Additional Responsibilities:
 - Assisting and guiding undergraduate students with their final-year projects.
 - Occasionally serving as an invigilator for undergraduate term examinations.
- **Computer Science Teacher** 09/2023 - 02/2024
Canadian Trillinium International School | [LOR by Course Coordinator] Dhaka, Bangladesh
Teaching Area:
 - *Programming in Python* (Grade 9)
 - *Programming in C++* (Grade 10)
 - *Cyber Security & Technical Support* (Grade 11 & 12)
- **Research Assistant** 09/2023 - 01/2024
Artificial Intelligence Research and Innovation Lab (AIRIL) Dhaka, Bangladesh
- **Cloud Computing Trainee** 06/2023 - 09/2023
eTech Solution Ltd. | [Completion Certificate] Dhaka, Bangladesh

EDUCATION

- **Master of Computing (by research)** 2024 - 2026
Multimedia University Melaka, Malaysia
 - Thesis: Hybrid Privacy-Preserving Federated Learning Framework with Homomorphic Encryption for Secure Data Sharing
[Thesis submission acknowledgement] [Title Approval Letter]
- **Bachelor of Science in Computer Science and Engineering (CSE)** 2020 - 2023
American International University-Bangladesh (AIUB) Dhaka, Bangladesh
 - **CGPA: 3.94 out of 4.00** [Magna Cum Laude, Dean's List Honor - Five Times] | [Certificate] [Transcripts]
 - Thesis: Innovative Techniques of Data Sharing and Security for Anomaly Detection Using Novel Machine Learning Techniques
- **Higher Secondary School Certificate (HSC)** 2016 - 2018
Sreemangal Government College Sreemangal, Bangladesh
 - GPA: 4.83 out of 5 | [Certificates & Transcripts]

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, U=UNDER PUBLICATION(ALREADY ACCEPTED), S=IN SUBMISSION(UNDER PEER REVIEW)

- [J.1] **A comprehensive Review of Cryptographic Techniques in Federated Learning for Secure Data Sharing and Applications.** *IEEE Access*, doi: 10.1109/ACCESS.2025.3593953. [Scopus Q1].
- [J.2] **BiLSTM-LIME: Integrating NLP and Advanced Machine Learning Models for Fake News Detection .** *Discover Artificial Intelligence*, doi.org/10.1007/s44163-026-00852-w. [Scopus Q1].
- [J.3] **Preventing Student's Mental Health Problems with the Help of Data Mining.** *International Journal of Computing*, vol. 23, no. 1, pp. 101-108, Oct. 2024, doi: 10.47839/ijc.23.1.3441 [Scopus Q3].
-
- [C.1] **Enhancing DDoS Attack Detection Through Hybrid Ensemble Machine Learning Technique and Explainable AI.** In *The 9th International Conference on Innovation in Artificial Intelligence (ICIAI 2025)*, *Lecture Notes in Electrical Engineering*, vol 1458. Springer, Singapore. doi: https://doi.org/10.1007/978-981-95-0887-7_3 [Scopus Q4].
- [C.2] **Privacy-Preserving Federated Learning for MRI-Based Brain Tumour Detection Using Xception CNN.** In *2026 International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML)*, Tokyo, Japan, 2026, pp. 124-129, doi: 10.1109/AAIML67890.2026.11498205. [Scopus Indexed].
- [C.3] **Fake news detection: An introduction of hybrid machine learning model outperforming conventional classifiers.** *AIP Conf. Proc.*, (Vol. 3462, No. 1, p. 060016). AIP Publishing LLC., doi: 10.1063/5.0312874 [Scopus and WoS Indexed].
- [C.4] **Fusion of machine learning techniques for robust anomaly detection in cybersecurity during data sharing.** *AIP Conf. Proc.*, (Vol. 3462, No. 1, p. 060008). AIP Publishing LLC., doi: 10.1063/5.0312873 [Scopus and WoS Indexed].
- [C.5] **Towards Precision Public Health: Advancing Obesity Prediction with Machine Learning Algorithms.** In *The IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT 2025)*, Bali, Indonesia, 2025, pp. 826-832, doi: 10.1109/IAICT65714.2025.11100376. [Scopus Indexed].
- [C.6] **From Data to Diagnosis: Applying Machine Learning for Reliable Heart Disease Prediction.** In *2025 The 5th Asian Conference on Information Engineering (ACIE 2025)*, Phuket, Thailand, 2025, pp. 50-56, doi: 10.1109/ACIE64499.2025.00015. [Scopus Indexed].
- [C.7] **Efficient American Sign Language Recognition using ResNet-50.** In *The 11th International Conference on Engineering, Applied Sciences and Technology (ICEAST 2025)*, Phuket, Thailand, 2025, pp. 152-158, doi: 10.1109/ICEAST64767.2025.11088186. [Scopus Indexed].
- [C.8] **From Filtering to Features: A Machine Learning Approach for Stress Detection.** In *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 3717-3722, doi: 10.1109/ICCIT68739.2025.11491727. [Scopus Indexed].
- [C.9] **Identification of Human Movement Through a Novel Machine Learning Approach.** In *2023 Innovations in Power and Advanced Computing Technologies (i-PACT)*, KL, Malaysia, 2023, pp. 1-5. [Scopus Indexed]
- [C.10] **Advancements in human movement monitoring: A comprehensive approach leveraging deep learning and machine learning techniques.** In *AIP Conference Proceedings*, Vol. 3367, No. 1, p. 020001, AIP Publishing LLC. [Scopus & WoS Indexed]
-
- [U.1] **A Federated Learning Framework for Brain Tumour Detection Using Xception CNN with CKKS-Based Secure Aggregation.** In *2026 14th International Conference on Information and Communication Technology (ICoICT)*, Bandung, Indonesia.
- [U.2] **PryML: Privacy-Preserving Federated Learning under Data Heterogeneity using CKKS Encryption.** In *2026 14th International Conference on Information and Communication Technology (ICoICT)*, Bandung, Indonesia.
- [U.3] **Secure Federated Learning Framework for ICU Mortality Prediction Using Hybrid XGB-NN.** In *The Multimedia University Engineering Conference 2026 (MECON 2026)*, Cyberjaya, Malaysia.
- [U.4] **A Privacy-Preserving Approach to Diabetes Diagnosis Using Artificial Neural Networks and Federated Learning Integrating Homomorphic Encryption.** In *The 5th International Conference on Computer, Information Technology and Intelligent Computing (CITIC 2025)*, Cyberjaya, Malaysia.
-
- [S.1] **Privacy-Preserving Federated Learning with Quantised, Encrypted and Masked Aggregation: A Case Study on ICU Mortality Prediction.** Submitted and under review at *IEEE Access*. [Scopus Q1].
- [S.2] **Computation-Aware Federated Optimisation with Quantised Secure Aggregation for Brain Tumour Classification.** Submitted and under review at *Discover Applied Sciences* . [Scopus Q1].
- [S.3] **Recent Advances in Privacy-Preserving Federated Learning for Financial Applications: A Systematic Review.** Submitted and under review at *Discover Artificial Intelligence* . [Scopus Q1].

PARTICIPATION AND CERTIFICATIONS

- **Presenter at International Conferences** 2024-2026
- 2026 International Conference on Advances in Artificial Intelligence and Machine Learning (AAIML 2026), **Tokyo, Japan** \ [Certificate]
 - The Multimedia University Engineering Conference 2026 (MECON 2026), **Cyberjaya, Malaysia** \ [Certificate]
 - The 5th International Conference on Computer, Information Technology and Intelligent Computing (CITIC 2025), **Cyberjaya, Malaysia** \ [Certificate]
 - The 2025 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT'2025), **Bali, Indonesia** \ [Certificate]
 - 2025 the 9th International Conference on Innovation in Artificial Intelligence (ICIAI 2025), **NTU, Singapore** \ [Certificate]
 - The 11th International Conference on Engineering, Applied Sciences and Technology (ICEAST 2025), **Thailand** \ [Certificate]
 - The 5th Asian Conference on Information Engineering (ACIE 2025) , **Phuket, Thailand** \ [Certificate]
 - 3rd International Conference on Big Data, IoT and Machine Learning (BIM 2025) , **Dhaka, Bangladesh** \ [Certificate]
 - 5th International Conference on Mathematics, Engineering & Industrial Applications (ICoMEIA) , **Langkawi, Malaysia** \ [Certificate]
- **Champion of the Three-Minute Thesis (3MT) competition 2026 (Master Level - MMU)** [Certificate] 2026
- **Gold Award (two) - iNVENTX - an international innovation exhibition 2025** 2025
- Winner of two research posters I presented at iNVENTX 2025. These researches are focused on DDoS attack detection using ML and privacy-preserving diagnosis in FL integrating homomorphic encryption, respectively. [Certificate 1], [Certificate 2]*
- **1st Runner-Up in Infineon-MMU Innovative Research Poster Competition 2025 (IT-Category)** [Certificate] 2025
- **Finalist - Three-Minute Thesis (3MT) competition 2025 (Master Level - MMU)** [Certificate] 2025
- **Gold Award - iNVENTX - an international innovation exhibition 2024** 2024
- Winner of a research poster I presented at iNVENTX 2024. The research focused on predicting human choice preferences in the chatbot arena. [Certificate]*

- **CNN Architectures in Industrial Applications** 2024
A five-day long training by AI Nusantara in physical mode, ending with a capstone project which is published in IEEE Explore | [Certificate]
- **National Research Fair-2023, CUET, Bangladesh** 2023
Participated with a selected research article titled "Revolutionizing Water Quality Assessment in Bangladesh: A Novel Hybrid Model Outperforming Conventional Classifiers." | [Certificate]
- **DUET TechFest 2k23** 2023
Winner of the segment of the Idea & Project Exhibition, project titled: Sign Language to Speech Converter. | [Certificate] | [Trophy]
- **Orbitax SUST SWE Technovent 2023 (Brain Station 23 Hackathon)** 2023
Successfully passed the first round and participated in the final round at SUST campus for a 36-hour non-stop onsite hackathon. | [Certificate]
- **Cisco IT Essentials** 2020
Successfully achieved the student level credential by completing the IT Essentials course. | [Certificate]

TECHNICAL SKILLS

Programming Languages:	C, C++, Java, C#, Python, JavaScript, Php, Rust
Data Science Tools:	PyTorch, TensorFlow, NumPy, Scikit-Learn, Matplotlib, OpenCV
Web Development:	FrontEnd - HTML5, CSS3, Bootstrap5, JavaScript BackEnd - ASP.NET
Software Testing tool:	Selenium
Database Management:	MySQL, MS SQL, PL SQL
Miscellaneous:	Git, L ^A T _E X

HONORS AND AWARDS

- **Magna Cum Laude (Silver Medal)** | [Qualifications & Criteria] | [Certificate] 2025
22nd Convocation, American International University-Bangladesh (AIUB)
- **High Achiever Research Scholarship for Postgraduate Programs** | [Confirmation Letter] 2024
Multimedia University, Melaka, Malaysia
- **FST Dean's List Honor** | [Certificates] 2023
American International University-Bangladesh | Awarded for academic excellence. [5x for Five Consecutive Academic Sessions]
- **Emerging Leader Award 2024** | [Certificate] 2024
DMH Foundation | Based on the top-performing academic excellence of undergraduate programs.
- **AIUB Academic Excellence Scholarship** 2021-2023
American International University-Bangladesh | Based on academic results. (Till Last Semester)

EXTRA-CURRICULAR ACTIVITIES

- Vice President of Postgraduate Society, MMU (Term 2026)
- Campus Ambassador [Team Lead] at **ICT Olympiad Bangladesh 2022** (Season 1) | [Certificate]
- Volunteer at **Bangladesh Red Crescent Society** | [LOR By Chief of the Training & Co-curricular Department]

REFERENCES

- | | | |
|---|--|--|
| 1. Prof. Ts. Dr. Heng Swee Huay
Professor, Faculty of Information Science and Technology, Multimedia University, Malaysia
Email: shheng@mmu.edu.my | 2. Ass. Prof. Dr. Didik Sudyana
Assistant Professor, Computer and Network Center, National Cheng Kung University (NCKU), Taiwan
Email: didik@gs.ncku.edu.tw | 3. Prof. Dr. Khandaker Tabin Hasan
Professor & Head of Graduate Program, American International University-Bangladesh (AIUB), Dhaka
Email: tabin@aiub.edu |
|---|--|--|